

Praneet Gogoi

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EDUCATION

Royal Global University

Bachelor of Technology in Computer Science | CGPA: 7.0

Guwahati, Assam

June 2026

Kendriya Vidyalaya

Class XII: 88% | Class X: 86%

May 2021

TECHNICAL SKILLS

Languages: Python, SQL, Shell

Visualization & BI: Plotly, Seaborn, Matplotlib, Power BI, Looker Studio

Web & APIs: Flask, FastAPI (basics), REST APIs, HTML/CSS/JS

Cloud & DevOps: Supabase (cloud PostgreSQL), Vercel (cloud deployment), Git, GitHub

Certifications: Google Data Analytics Professional, HackerRank Python

PROFESSIONAL EXPERIENCE

Indian Oil Corporation Limited (IOCL)

AI/ML Intern

Guwahati, Assam

July 2025 – Aug 2025

- Built ML models (Scikit-learn, TensorFlow) to analyze refinery operational datasets, reducing analytical task time by **40%** through automated Python data workflows.
- Developed predictive maintenance models forecasting equipment failure windows from sensor data, improving scheduling efficiency for a facility processing **1M+ daily records**.
- Engineered 10+ domain features from industrial time-series data and applied ensemble classifiers to optimize energy management system parameters.

National Institute of Electronics & IT (NIELIT)

AI/ML Intern

Guwahati, Assam

July 2024 – Aug 2024

- Implemented ML models for regression, face detection, and image classification using TensorFlow and Keras, achieving **>85% accuracy** across three classification tasks.
- Built an end-to-end face detection pipeline with OpenCV and a CNN-based classifier evaluated on 2,000+ image samples.
- Strengthened understanding of hardware-software integration through semiconductor technology study and digital circuit simulation.

PROJECTS

IndicSignAI | *EfficientNetV2-S, Transformer, NLLB-200, T5, Flask, WebGL, MediaPipe* | [GitHub](#)

- Built a real-time bidirectional ISL recognition and translation system supporting **6 regional languages** (Hindi, Assamese, Meitei/Manipuri, Nepali, Dzongkha, Lepcha) for India's 63M deaf and hearing-impaired population; deployed across **5 educational and healthcare institutions** in Northeast India for 6 months.
- Designed a CNN-Transformer hybrid architecture (EfficientNetV2-S backbone + 4-layer Transformer encoder) achieving **92.3% top-1 accuracy** on a **178-class ISL dataset** (39,600 sequences) at **45 ms CPU inference** — outperforming 3D-ResNet50 (84.2%) and I3D Conv (88.1%) baselines while requiring no GPU.
- Integrated Meta NLLB-200 distilled translation engine with a custom T5-small Bengali-to-Meitei Mayek transliterator achieving **96.8% character-level accuracy** (7.8pp above Google Translate baseline); delivered the **first computational support for Lepcha script** in any deployed accessibility system.
- Six-month deployment produced **127,100 successful translations**, **91.2% healthcare communication success rate**, **23% improvement in deaf student comprehension**, and **87.2% monthly user retention** across 5 sites; WebGL avatar animation scored **5.8/7.0 naturalness** (vs. 6.1/7.0 human signer gold standard).

E-commerce Customer Churn Prediction | *Python, Scikit-learn, Pandas, PostgreSQL, Plotly, Vercel* | [GitHub](#)

- Built an end-to-end ML pipeline using Random Forest classifier to predict customer churn across **6,000 records** with **14 engineered behavioral features**; achieved AUC >0.82 on holdout test set.

- Designed production-ready PostgreSQL schema with a unified analytics view for direct BI tool integration; **cloud-deployed** interactive frontend on Vercel with live prediction API endpoint.
- Generated SHAP feature importance plots surfacing top churn drivers (tenure, complaint count, cashback amount) for business stakeholders.

NAS Insights — India Macro Data Warehouse | *PostgreSQL, Python, Pandas, SQLAlchemy, Supabase* | [GitHub](#)

- Designed a Star Schema data warehouse over MoSPI macroeconomic datasets spanning **30+ years** of GDP, GVA, and expenditure data; built Python ETL pipeline with Pandas and SQLAlchemy to ingest and normalize annual/quarterly records.
- Built pre-computed SQL views for economic KPIs (GDP growth rates, trade balance, quarterly seasonality) and stored procedures for dynamic Top-N industry rankings; **cloud-deployed on Supabase** (managed PostgreSQL).

Brain MRI Hybrid Deep Learning Model | *TensorFlow, Keras, CNN, Transfer Learning, OpenCV* | [GitHub](#)

- Built a hybrid deep learning model (EfficientNet CNN + ensemble ML classifiers) for brain tumor classification from MRI images; applied image augmentation and preprocessing on **medical imaging datasets with 4 tumor classes**.
- Achieved **>90% validation accuracy** using transfer learning fine-tuning; evaluated with confusion matrix and per-class F1 scores to address class imbalance.

Selected Data Analysis Projects | *Python, Pandas, Scikit-learn, Plotly, Seaborn, Jupyter* | [GitHub](#)

- **Flight Delay Prediction:** Delay classification and pattern analysis from airline records using Random Forest and XGBoost; identified top 5 delay-causing routes.
- **Loan Default Prediction:** End-to-end classification pipeline on financial datasets; engineered 8 behavioral features with 87% precision on minority class.
- **Sentiment Analysis (Comments):** NLP classification pipeline using TF-IDF + Logistic Regression on 50K+ user comments; deployed as interactive HTML dashboard.
- **Heart Health & Cancer Risk Analysis:** Multi-class risk classification on clinical datasets; SHAP-based feature attribution for interpretable medical insights.
- **Valorant Champions Analysis:** Esports analytics dashboard — agent pick rates, KDA distributions, map win rates — built with Plotly and hosted as interactive HTML.

RESEARCH

IndicSignAI: Real-Time Bidirectional ISL Recognition & Regional Language Translation

- **Authors:** Sadique Ahmed, Harsimran Singh, Praneet Gogoi | Supervisor: Dr. Nilakshi Deka, RSET, Royal Global University | Submitted to *Expert Systems With Applications* (Elsevier Q1).
- **Architecture:** EfficientNetV2-S CNN backbone (~21.5M params) + 4-layer multi-head Transformer encoder; trained on a self-collected **39,600-sequence, 178-class ISL dataset** gathered from 25 native ISL signers across Assam, Manipur, Sikkim, and Nagaland. Achieves **92.3% top-1 / 98.1% top-5 accuracy** at **45 ms CPU inference** (Intel i5-8250U, 8 GB RAM, no GPU).
- **Translation layer:** Meta NLLB-200 (600M distilled) for Hindi, Assamese, Nepali; custom T5-small fine-tuned on 5,000 Bengali–Meitei Mayek pairs (**96.8% char-level**, 92.3% word-level); Tibetan Unicode normaliser for Dzongkha; Nepali-pivot rule-based pipeline for Lepcha — the **first computational integration of Lepcha script** into any NLP or accessibility system.
- **Translation quality:** BLEU-4 scores of 42.7 (Hindi), 35.4 (Assamese), 34.5 (Dzongkha), 32.7 (Nepali), 29.6 (Meitei); average human fluency 4.0/5.0, adequacy 4.3/5.0 by native speaker evaluation.
- **Avatar animation:** Procedural WebGL avatar using quaternion SLERP interpolation renders genuine ISL motion at **60 FPS** in-browser with no pre-recorded video; naturalness score **5.8/7.0** vs. human signer gold standard of 6.1/7.0 (blind evaluation, $n = 30$).
- **Real-world deployment (6 months, 5 sites):** 127,100 successful translations | 91.2% healthcare communication success | 23% improvement in deaf student academic comprehension | 87.2% monthly user retention.